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1 (a) Give three structural features of a root hair cell and, for each, state how it helps water or ion uptake. [6]

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(b) Name the process by which water enters a root hair, and the process by which nitrate ions enter when their concentration is higher inside the cell than in the soil. [2]

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[Turn over]

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2 A chemical stops aerobic respiration in roots. Nitrate uptake slows; water uptake does not. Explain both observations. [4]

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[Turn over]

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3 (a) Name, in the correct order, the four tissues / cells that water passes through from soil water to a leaf mesophyll cell. [4]



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(b) Name the tissue that transports mineral ions from the root to the leaf. [1]

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[Turn over]

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4 (a) A leafy shoot is cut under water and stood in eosin. A transverse section of the stem is cut after 1 hour. Which tissue is stained? [1]

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(b) Explain why the shoot is cut under water. [1]

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(c) Explain why the dye moves up the stem. [1]

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[Turn over]

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5 Define transpiration. [2]

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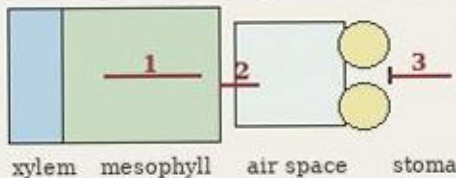
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[Turn over]

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6 Fig. 6.1 shows water moving through part of a transpiring leaf. For each of arrows 1, 2 and 3, name the process taking place and define it. [9]



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[Turn over]

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7 Explain how each of the following changes the rate of transpiration: (i) higher wind speed (ii) higher temperature (iii) higher humidity (iv) higher light intensity. [8]

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[Turn over]

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8 Leaves from two species transpire at different rates in the same environmental conditions. Suggest three structural differences in the lower surface of these leaves which might produce the difference. Give a reason for each. [6]

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[Turn over]

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9 (a) Explain how wilting occurs in a plant. [4]

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(b) A plant is pulled from the soil and replanted. It wilts for several days and then recovers. Explain both stages. [4]

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[Turn over]

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10 Bacteria reproduce inside a plant stem and block the xylem. The leaves wilt and become yellow. Explain both of these symptoms.

[5]

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[Turn over]

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11 Softwood cuttings have their lower leaves removed and are covered with a transparent bag until roots develop. Explain how each of these two steps helps to prevent wilting.

[4]

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[Turn over]

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12 Explain the mechanism by which water moves upwards in the xylem.

[5]

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[Turn over]

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13 What is the main cause of water moving up to the leaves in xylem vessels? State the correct idea and identify three incorrect ideas that are often given as options.

[2]

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14 (a) A potted plant is watered and the pot is placed in a plastic bag. Mass at 09:00 = 1276 g. Mass at 15:00 = 1252 g. Calculate the mean rate of transpiration over the six hours. Show your working. [2]

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(b) Explain why the pot was placed in a plastic bag. [1]

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[Turn over]

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15 State the combination of humidity, temperature, wind and light intensity that gives the highest rate of transpiration. [1]

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[Turn over]

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16 Transpiration is both necessary for plants and a problem for plants. Discuss this statement. [4]

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[Turn over]

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17 (a) Define translocation. [3]

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(b) Give one example of a source and one example of a sink. [2]

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(c) State one way in which the direction of translocation differs from transport in the xylem. [1]

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[Turn over]

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18 Aphids feed by piercing one plant tissue. Name this tissue and explain why it is more suitable for this purpose than xylem.

[3]

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[Turn over]

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19 When flowers are cut for display, the leaves are removed and the stems are placed in a sucrose solution. Explain the benefit of providing sucrose to the cut flowers.

[3]

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20 Fig. 20.1 shows a TS of a non-woody dicotyledonous stem. Fig. 20.2 shows a TS of a non-woody dicotyledonous root. For each figure, state the positions of the xylem, the phloem and the cortex.

[6]

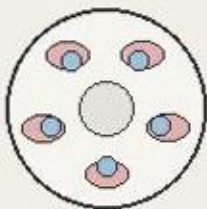


Fig. 20.1 stem TS

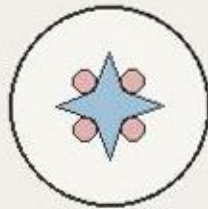


Fig. 20.2 root TS

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[Turn over]

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21 (a) State two functions of xylem vessels in a young stem. [2]

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(b) For each function, explain one way in which xylem structure is adapted for that function. [4]

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[Turn over]

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22 Relate each of the following features of xylem vessels to their function: (i) thick walls with lignin (ii) no cell contents (iii) cells joined end-to-end with no cross walls. [6]

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[Turn over]

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23 List three structural changes that occur in young plant cells as they develop into xylem tissue. [3]

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[Turn over]

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24 Describe two similarities and two differences in the structure or function of root hairs and villi. [4]

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[Turn over]

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25 Define excretion.

[3]

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[Turn over]

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26 Explain why egestion is not excretion.

[2]

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[Turn over]

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27 Which of the following is an example of excretion? Explain why the others are not. • release of a hormone into the blood • removal of carbon dioxide from the lungs • removal of undigested food from the digestive system • release of water from the sweat glands

[3]

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[Turn over]

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28 Carbon dioxide is a waste product of metabolism. State where it is produced and how it is excreted.

[2]

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[Turn over]

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29 (a) Name the organ in which urea is produced.

[1]

(b) State what urea is produced from.

[1]

(c) Explain why urea must be excreted.

[1]

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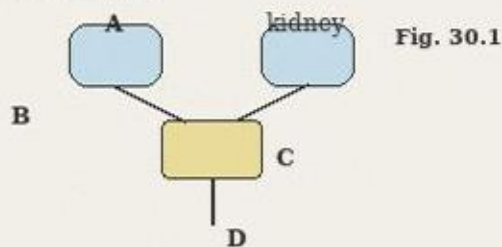
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30 Fig. 30.1 shows the human urinary system. Identify structures A to D and state the function of each.

[8]



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[Turn over]

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31 Describe two similarities or differences between the ureter and the urethra.

[4]

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[Turn over]

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32 Fig. 32.1 is a simplified diagram of a nephron. Identify the structures labelled P, Q and R. [3]

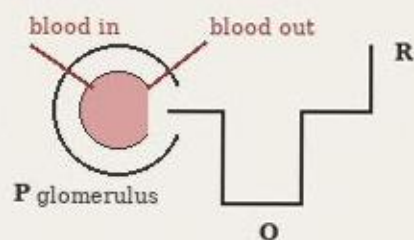


Fig. 32.1

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[Turn over]

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33 (a) Outline the role of the glomerulus. Name four substances filtered from the blood. [3]

(b) State what is reabsorbed by the nephron, and how much of each (all / some / most). [3]

(c) Name three components of the urine that is formed. [2]

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[Turn over]

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34 Fluid leaving the collecting duct of a healthy person contains glucose at concentration 0.00 and protein at concentration 0.00. Use your knowledge of the function of a nephron to explain each of these zeros. [5]

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[Turn over]

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35 The table shows amounts entering the kidneys in the blood and leaving in the urine in one day. water $150 \text{ dm}^3 \rightarrow 1.5 \text{ dm}^3$ glucose $150 \text{ g} \rightarrow 0 \text{ g}$ urea $50 \text{ g} \rightarrow 30 \text{ g}$. Calculate the percentage of each substance that is reabsorbed. [3]

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[Turn over]

5090/11/M/J/23

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36 Compare the blood in the renal artery with the blood in the renal vein with respect to urea, oxygen and carbon dioxide. [3]

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[Turn over]

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37 Which blood vessel has the highest concentration of urea, and which has the lowest: hepatic portal vein, hepatic vein, renal artery, renal vein? Explain your answers. [3]

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[Turn over]

5090/21/O/N/25

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38 A person eats more protein than usual. Explain why the concentration of urea in the urine rises. [3]

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[Turn over]

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39 A person drinks a large volume of water. State and explain two changes in the urine.

[3]

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[Turn over]

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40 Which of the following would not be present in the urine of a healthy person with a normal diet: glucose, salts, urea, water?

[1]

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[Turn over]

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41 (a) Name the process by which amino acids pass from the ileum into the blood. [1]

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(b) Name the process of converting amino acids to proteins in the liver. [1]

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(c) Name the blood vessel that carries amino acids from the small intestine to the liver. [1]

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(d) Name the process in the liver that produces urea from excess amino acids. [1]

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[Turn over]

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42 Define deamination. [2]

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[Turn over]

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43 The diagram of an amino acid molecule is not shown. Which part of the molecule is removed and used to make urea: the -NH_2 group, the R group, or the -COOH group? [1]

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[Turn over]

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44 Outline the path of the nitrogen from an absorbed amino acid until it leaves the body as urea. [6]

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[Turn over]

Question	Answer	Marks / guidance
1	(a) long/projection/finger-like → large SA ; large SA → absorb water and ions ; thin wall → short distance ; (partially permeable) membrane → osmosis of water / AT of ions ; mitochondria/living → ATP for ions ; vacuole → store / turgor ; [max 6] (b) water: osmosis (A diffusion) ; ions: active transport ; [2] R water by AT ; R ions by osmosis ; R partially permeable cell wall	
2	ions need AT / against gradient ; AT needs energy from respiration ; chemical stops respiration → slower ion uptake ; water by osmosis / does not need energy ; [4]	
3	(a) root hair (cell) ; (root) cortex (cell) ; xylem (vessel) ; palisade/spongy mesophyll (cell) ; this order [4] (b) xylem ; [1] R phloem	
4	(a) xylem ; (b) prevent air lock / air bubbles in xylem ; (c) transpiration pull / evaporation from leaves draws water + dye up ; [3]	
5	loss of water ; as vapour / evaporation ; from leaves ; through stomata ; [max 2] R evaporation from the epidermis as the main cause of the pull	
6	1 osmosis ; water higher → lower water potential across a partially permeable membrane ; 2 evaporation ; liquid → water vapour (from mesophyll wall/film) ; 3 diffusion ; vapour high → lower concentration / out through stomata ; [9]	
7	wind ↑: vapour blown away ; humidity around leaf falls ; steeper diffusion gradient ; temp ↑: more KE ; faster evaporation ; humidity ↑ rate ↓: smaller vapour-concentration difference ; shallower gradient ; R stomata as the humidity explanation ; light ↑: stomata open/wider ; [8]	
8	leaf SA ; stomata number/density/size ; stomata upper vs lower ; cuticle thickness ; sunken stomata/hairs/rolled leaves trap humid air ; [max 6 — feature + reason]	
9	(a) water lost (transpiration) faster than gained ; cells lose turgor / become flaccid ; no longer support plant ; [4] Prefer flaccid (plasmolysed R on an older wilting MS). (b) wilt: root hairs damaged ; less water absorbed ; recover: hairs regrow ; cells turgid again ; [4]	
10	water lost cannot be replaced / cannot pass up stem ; lose turgor → wilt ; Mg cannot reach leaves ; less chlorophyll → yellow ; less photosynthesis ; [5]	
11	remove leaves: less SA/fewer stomata → less transpiration ; bag: higher humidity ; smaller diffusion gradient ; less water loss ; [4]	
12	evaporates/transpires from leaves ; lowers leaf water potential ; water up xylem ; transpiration pull ; forces of attraction between water molecules / column held together ; [max 5] R root pressure/capillarity as the main mechanism ; R active transport	
13	correct: evaporation from the walls of the mesophyll cells ; wrong: active transport ; evaporation from the epidermis ; use of water in photosynthesis ; [2]	
14	(a) $1276 - 1252 = 24 \text{ g in } 6 \text{ h} \rightarrow 4 \text{ g/h}$; [2] (b) so mass loss is leaf transpiration, not soil evaporation ; [1]	
15	low humidity + high temperature + wind + high light/day ; [1]	
16	necessary: water for photosynthesis ; ions to leaves ; turgor ; cools leaf ; problem: too much loss → wilt/death ; close stomata → less CO ₂ → slower photosynthesis ; [4]	
17	(a) sucrose and amino acids ; in phloem ; source (produce/release) → sink (use/store) ; in solution ; [max 3] R glucose/food (b) source = leaf (or storage releasing) ; sink = root/meristem/fruit ; [2] (c) phloem bidirectional ; xylem up only ; [1]	
18	phloem ; sucrose + amino acids ; in solution ; not lignified/softer ; [max 3]	
19	cut flowers cannot photosynthesise ; sucrose in phloem ; respiration/energy ; last longer ; [max 3]	
20	STEM: bundles in a ring towards outside ; xylem inner ; phloem outer ; cortex between epidermis and ring ; ROOT: xylem in centre (star) ; phloem between arms ; wide cortex ; [6]	
21	(a) transport water and mineral ions ; support ; [2] R sugars (b) transport ↔ hollow / no cytoplasm / no end walls / continuous tube ; support ↔ lignified/thick walls ; pair them ; [4]	

Question	Answer	Marks / guidance
22	(i) lignin → support / stop collapse ; (ii) no contents → hollow / free flow ; (iii) no cross walls → continuous tube / uninterrupted column ; [6]	
23	elongation ; loss of contents / death / hollow ; loss of end walls ; add lignin ; [max 3]	
24	same: elongated/large SA ; absorption ; AT/diffusion ; water/ions ; diff: hair = one cell + wall, from soil ; villus = multicellular + vessels, from gut ; [4]	
25	removal from the organism ; of toxic/poisonous materials ; and waste products of metabolism ; [3] R harmful ; R salts as the definition	
26	egestion = undigested food / fibre/cellulose ; not from metabolism ; [2]	
27	excretion = CO ₂ from lungs ; hormone = secretion ; faeces = egestion ; sweat = cooling/homeostasis ; [3]	
28	waste of (aerobic) respiration ; excreted through lungs / exhaled / blood → alveoli ; [2]	
29	(a) liver ; (b) breakdown of excess amino acids ; (c) urea is toxic ; [3] R kidney produces urea	
30	A kidney: removes urea + excess salts + water from blood as urine ; B ureter: urine kidney → bladder ; C bladder: stores urine ; D urethra: urine bladder → out ; R 'just urea' ; male urethra also sperm ; [8]	
31	both carry urine ; ureter kidney → bladder ; urethra bladder → out ; male urethra also sperm ; [4]	
32	P Bowman's capsule ; Q loop of Henle ; R collecting duct ; [3]	
33	(a) filtration ; water, glucose, urea, ions ; into Bowman's ; [3] (b) all glucose ; some ions ; most water ; back into blood ; [3] (c) urea ; excess water ; excess ions ; [2]	
34	glucose: filtered into Bowman's ; then all reabsorbed ; protein: too large ; not filtered ; [5]	
35	water 99% ; glucose 100% ; urea 40% ; [3] R quoting amount (148.5) as a %	
36	renal vein: less urea ; less oxygen ; more carbon dioxide ; [3]	
37	highest = hepatic vein (liver just made it) ; lowest = renal vein (kidney just removed it) ; [3]	
38	more amino acids deaminated (liver) ; more urea formed ; more urea filtered into urine ; [3]	
39	higher urine volume ; lower ion concentration / more dilute ; homeostasis of water in blood ; [3]	
40	glucose ; [1]	
41	(a) absorption ; (b) assimilation ; (c) hepatic portal vein ; (d) deamination ; [4]	
42	removal of the nitrogen-containing part (-NH ₂) of amino acids ; formation of urea ; [2]	
43	the nitrogen-containing / -NH ₂ group ; [1]	
44	absorbed in villi ; hepatic portal vein ; liver ; deamination ; urea in renal artery ; kidney filters ; urine ; ureter → bladder → urethra ; [max 6]	